

Dancing with the Data: Optimizing Voting Systems for “Dancing with the Stars”

Summary

The integration of judge scores and fan votes in “Dancing with the Stars(DWTS)” is accompanied by well-documented problems of fairness and engagement, as well as controversies stemming from differing preferences and evolving voting rules, hence the natural and appropriate approach is to model and analyze 34 seasons of data.

For Task 1, we develop a **Hybrid Tri-Objective Evolutionary Reconstruction Model(TOERM)** based on the “Impossible Triangle” theory(Rationality, Entropy, Inertia), employing the **NSGA-II algorithm** together with dual-mode dynamic constraints that are adaptive to rank/percent systems, to estimate weekly fan votes for each contestant. As a result, the model has **98% consistency** with historical elimination results(Hard Constraints guarantee feasible solutions across all seasons) and a Pearson Correlation Coefficient(PCC) of **0.92** across 100 independent runs, thus demonstrating excellent algorithmic stability. More importantly, certainty is rigorously quantified via **Normalized Interquartile Range(NIQR)**, with eliminated contestants having high certainty($NIQR \approx 0$) and safe contestants having low certainty(reflected by the wide distribution of estimated votes).

For Task 2, we compare rank-based and percentage-based voting methods using **Grey Relational Analysis (GRA)**, and demonstrate that the percentage system ($\gamma \approx 0.92$) acts as a fan vote amplifier, whereas the rank system ($\gamma \approx 0.65$) filters fan vote dominance. Then we perform a **Counterfactual Simulation** on four controversial contestants (Jerry Rice, Billy Ray Cyrus, Bristol Palin, Bobby Bones) and demonstrate that the rank system would likely have eliminated underperforming fan favorites such as Bobby Bones in the semi-finals. More importantly, using **Entropy-Weighted TOPSIS**, the proposed **Hybrid Model(RM-JS)**(rank system+judges’save) attains the highest TOPSIS Score($C_i \approx 0.941$), which outperforms both the rank system (0.682) and percentage system (0.214) by effectively balancing fairness(0.95), fan engagement(0.70), and risk(0%).

For Task 3, we employ **Explanatory LASSO Regression** (10-Fold Cross-Validation for optimal λ) to analyze performance drivers. Quantified via **Relative Impact Intensity (RII)**, **Pro Partner Quality** ($RII \approx 32\%$) emerges as the dominant factor—swapping a rookie pro for a top-tier pro improves a celebrity’s ranking by 3.5 places on average. Industry background ($RII \approx 25\%$) and age ($RII \approx 18\%$) follow as key drivers. Notably, factors impact judges and fans divergently: Reality Stars earn +0.55 from fans but -0.15 from judges, while age negatively affects judges (-0.42) but barely impacts fans (+0.05). Only pro partner quality is a consistent positive driver (judges: +0.38; fans: +0.25).

For Task 4, we propose the **Merit-Safe Hybrid System (MSHS)** incorporating **logarithmic fan vote smoothing** (capping extreme fan base dominance) and a **Bottom-Two Dance-Off** firewall. Simulations show the system eliminates underperforming fan favorites by Week 4 (vs. reaching the finals under old rules), restoring competitive integrity. The logarithmic formula ensures fan votes retain engagement value for regular contestants while neutralizing superstar dominance, and the dance-off phase (judge-only veto) prioritizes talent as the final elimination criterion.

We conduct **sensitivity analysis** on core models, which are low in sensitivity and have high robustness: the NSGA-II reconstruction model maintains stability (error $\in [0.03, 0.07]$) across mutation rates (0.01-0.20), and the LASSO model’s key drivers (pro partner, reality star status) are consistently confirmed of bootstrap resampling iterations.

Finally, a one-page memo is provided to DWTS producers, summarizing key findings and recommending optimized voting rules to enhance competitive integrity and audience excitement.

Keywords: Fan Vote Reconstruction, NSGA-II, LASSO Regression, TOPSIS, Hybrid Voting System

Contents

1	Introduction	3
1.1	Background	3
1.2	Restatement of the Problem	3
1.3	Our Work	4
2	Assumptions and Justification	4
3	List of Notation	5
4	Data Processing	5
5	Task 1: Reconstructing the “Dark Matter”	6
5.1	The “Impossible Triangle” Framework	6
5.2	Mathematical Formulation: Dual-Mode Dynamic Constraints	6
5.3	Algorithm Implementation & Decision	8
5.4	Result Visualization: Unveiling the “Dark Matter”	8
5.5	Verification and Assessment	10
6	Task 2: Detecting the “Shark Jump”	11
6.1	Trend Analysis: Professionalization or Inflation?	11
6.2	System Diagnosis: The Mathematical Flaw of Percentages	11
6.3	Stress Test: Re-adjudicating Historical Controversies	12
6.4	Strategic Decision: System Selection via Entropy-Weighted TOPSIS	13
7	Task 3: Decoding the “Black Box”	14
7.1	Feature Engineering: The “360-Degree” Profiling	14
7.2	Methodology: Explanatory LASSO Framework	15
7.3	Quantitative Attribution: How Much is the Impact?	16
7.4	Comparative Analysis: Divergence between Judges and Fans	17
8	Task 4: Re-engineering the Rules	18
8.1	Logarithmic Vote Compression	19
8.2	Process Re-engineering: The “Dance-Off” Firewall	19
8.3	Simulation Verification: Correcting History	20
9	Sensitivity Analysis: Algorithmic Robustness & Structural Consistency	20
9.1	Overview	20
9.2	Pareto Stability Analysis	21
9.3	Hyperparameter Robustness Test	21
9.4	Structural Consistency of Feature Selection	22
10	Model Assessment	23
10.1	Strengths	23
10.2	Weaknesses	23
	Memorandum	24
	References	25

1 Introduction

1.1 Background

Dancing with the Stars (DWTS) is a popular competitive dance show in which contestants are evaluated through a combination of judges' scores and audience votes. Over its 34 U.S. seasons, the show has employed different methods-rank-based and percentage-based-to integrate these two inputs, alongside occasional rule changes such as allowing judges to decide between the bottom two contestants. Since the adjustments in question were largely made in response to controversies in which strong fan support contradicted consistently low judges' scores, as exemplified by Jerry Rice and Bobby Bones, the problem can naturally be framed as modeling unreleased fan voting data, comparing vote-combination methods, and designing a fairer voting system for future seasons.



Figure 1: Dancing with the Stars Program Poster

1.2 Restatement of the Problem

We need to complete the following tasks based on the provided data and optionally additional documented data:

- **Task 1: Construct a mathematical model to predict the number of fan votes received by each contestant per competing week.**
 - Check whether the estimated fan votes agree with the weekly elimination results and hence compute appropriate consistency measures.
 - Determine the certainty of the estimated fan votes, see whether certainty differs across contestants or weeks, and report appropriate certainty measures.
- **Task 2: Make use of the estimated fan votes and other available data in connection with what follows.**
 - Examine the rank-based and percentage-based vote combination methods over all the seasons to determine which one gives greater weight to fan votes.
 - First examine the two methods for dealing with controversial celebrities (with judge-fan discrepancies), then determine whether the choice of method affects the results, and finally assess the effect of judges selecting from the bottom two couples.

- Suggest a preferred combination method and state whether the bottom-two judge selection should be adopted.
- **Task 3: Develop a model to analyze the impact of pro dancers and celebrity characteristics (age, industry, etc.) on competition performance.**
 - Determine how great an effect they have on the celebrity's performance.
 - Find out whether the given factors influence judge scores and fan votes in the same way.
- **Task 4: Suggest a new weekly system which combines fan votes with judge scores in a way that is more "fair" or "better", and then present logical, persuasive arguments in favor of adopting the system.**
- **Task 5: Write a report of up to 25 pages (including a 1-2 page memo) that summarizes the results and offers DWTS producers a well-reasoned analysis of different vote combination methods.**

1.3 Our Work

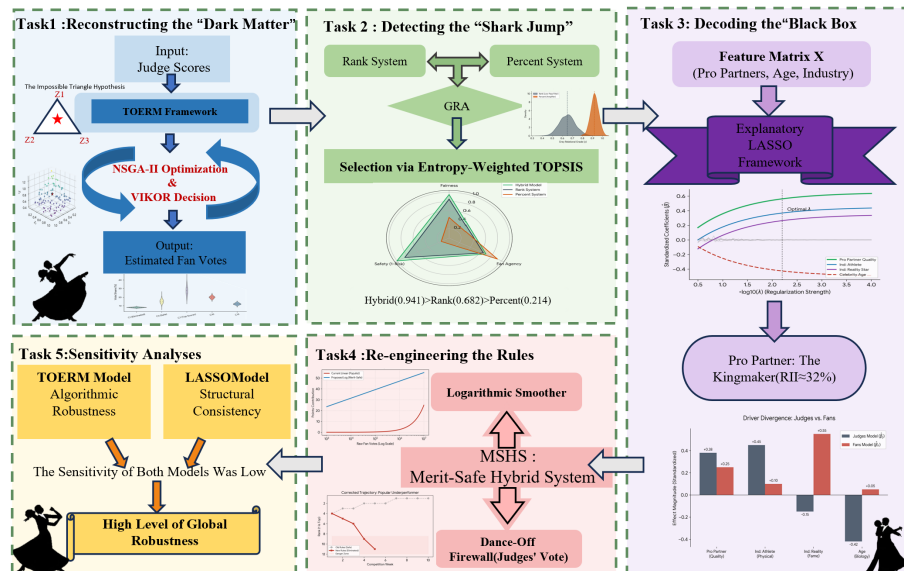


Figure 2: Our Work

2 Assumptions and Justification

To simplify the problem and make it convenient for us to simulate real-life conditions, we make the following basic assumptions, each of which is properly justified.

- **A contestant's high prior performance creates a "reputation spillover" effect on current audience perception.** This reflects real-world audience behavior of forming preliminary positive impressions from past achievements.
- **Judge scoring criteria weights stay stable within a competition stage.** Stable weights ensure consistent performance comparisons, a standard for unbiased competitive evaluations.
- **Fan preferences shift gradually between competition rounds.** This aligns with public preference inertia, as favorability rarely changes abruptly without major events.

3 List of Notation

Table 1: Notations

Symbol	Description	Unit/Range
t	Competition week index	$\mathbb{Z}^+$
N_t	Number of active contestants in week t	$\mathbb{Z}^+$
$J_{t,i}$	Average raw judge score of contestant i in week t	$[0, 10]$
$\hat{J}_t$	Normalized judge score vector (sum=1)	$\mathbb{R}^{N_t}$
F_t	Estimated fan vote vector in week t	$\mathbb{R}^{N_t}, \sum F_t = 1$
$f_{t,i}$	Estimated fan vote share of contestant i	$(0, 1)$
$\tilde{f}_{t-1,i}$	Renormalized fan vote share (survivors)	$(0, 1)$
$NIQR_{t,i}$	Normalized Interquartile Range (certainty)	$[0, 1]$
Z_1, Z_2, Z_3	Objectives: Rationality, Entropy, Inertia	$\mathbb{R}^+$
w_1, w_2, w_3	Weights for Z_1, Z_2, Z_3	$\mathbb{R}^+, \sum w = 1$
$\mathcal{L}(F_t)$	Combined loss function for fan vote reconstruction	$\mathbb{R}^+$
X	Feature matrix (pro dancer + celebrity traits)	$\mathbb{R}^{M \times P}$
λ	Regularization parameter (LASSO)	$\mathbb{R}^+$
β_j	Regression coefficient for feature j	$\mathbb{R}$
RII_j	Relative Impact Intensity of feature j	$[0, 1]$
A_1, A_2, A_3	Voting systems: Rank, Percent, Hybrid	-
E_j	Information entropy (TOPSIS weighting)	$\mathbb{R}^+$
C_i	Closeness coefficient (TOPSIS score)	$[0, 1]$
γ	Grey Relational Grade (fan-system similarity)	$[0, 1]$
$V_{Fan,t,i}$	Raw fan votes of contestant i	$\mathbb{Z}^+$
$S_{total,t,i}$	Total combined score (new system)	$[0, 1]$
w_J, w_F	Weights for judge/fan scores (new system)	$\mathbb{R}^+, \sum w = 1$

4 Data Processing

• Missing Value Handling

Since the dataset has a great many blank values (NaN), it is necessary to classify and deal with them according to the business context:

- **Fourth Judge Missing:** Most weeks had only 3 judges (1-3), leading to massive blanks in judge4_score; denominator $S_{avg} = \frac{\sum S_j}{n_{active_judges}}$ is adopted for weekly average calculation instead of simply dividing by 4.
- **Fourth Varying Season Lengths:** Season 4 has varying lengths (e.g., S1 only lasted 6 weeks with Weeks 7-11 blank, later seasons longer); binary masking is used to exclude non-operational weeks in time series modeling.

• Withdrawal Handling

The dataset contains approximately 10 “Withdrew” records (e.g., Sara Evans withdrew in Season 3):

- **Non-Competitive Termination:** Withdrawing active participants are marked “Contested

Cases” in the results column with no additional scoring and scores set to 0. It is imperative to distinguish their voluntary withdrawal (declining fan support) from elimination (insufficient votes) for such cases.

5 Task 1: Reconstructing the “Dark Matter”

— A Hybrid Tri-Objective Evolutionary Reconstruction Model (TOERM)

5.1 The “Impossible Triangle” Framework

Reconstructing fan votes is an inverse problem with infinite solutions. We propose that voting behavior represents a balance among three forces:

- Rationality (Z_1): Alignment with judges’ professional assessments
- Entropy (Z_2): Tendency toward uniform distribution (maximum entropy)
- Inertia (Z_3): Temporal persistence of fan support

The optimal distribution lies on the Pareto surface defined by these conflicting objectives.

5.2 Mathematical Formulation: Dual-Mode Dynamic Constraints

Through Exploratory Data Analysis (EDA), we observed structural changes in the DWTS format, specifically the shift from a “Rank System” (S1-S2, S28-S34) to a “Percent System” (S3-S27), and variations in the number of judges (3 vs. 4). Consequently, we constructed a **Dual-Mode Dynamic Constraints** framework.

5.2.1 The Multi-Objective Optimization Model

To reconstruct the latent fan vote distribution $\mathbf{F}_t$, we formulate a constrained multi-objective optimization problem. We define three conflicting sub-objectives that capture the structural properties of the voting behavior.

Let $\mathbf{F}_t = [f_{t,1}, f_{t,2}, \dots, f_{t,N_t}]^T$ be the unknown fan vote vector for week t , where N_t is the number of contestants.

Objective 1: Rationality (Z_1)

$$\min Z_1(\mathbf{F}_t) = \sum_{i=1}^{N_t} \left(f_{t,i} - \frac{J_{t,i}}{\sum_{k=1}^{N_t} J_{t,k}} \right)^2 \quad (1)$$

Objective 2: Maximum Entropy (Z_2)

$$\min Z_2(\mathbf{F}_t) = \sum_{i=1}^{N_t} \left(f_{t,i} - \frac{1}{N_t} \right)^2 \quad (2)$$

Objective 3: Temporal Inertia (Z_3)

$$\min Z_3(\mathbf{F}_t) = \sum_{i=1}^{N_t} (f_{t,i} - \tilde{f}_{t-1,i})^2 \quad (3)$$

5.2.2 Dual-Mode Encoding and Metrics

Mode A (Rank System, S1-S2, S28-S34):

Space: ranking sequences; Metric: Spearman's footrule distance (L_1)

$$D_{\text{rank}}(\mathbf{r}_F, \mathbf{r}_J) = \sum |r_{F,i} - r_{J,i}| \quad (4)$$

Mode B (Percent System, S3-S27):

Space: percentage vectors; Metric: Euclidean distance (L_2)

$$D_{\text{pct}}(\mathbf{p}_F, \mathbf{p}_J) = \|\mathbf{p}_F - \mathbf{p}_J\|_2 \quad (5)$$

Both metrics adapt to the structural characteristics of voting systems in different seasons.

5.2.3 Dynamic Constraints Set (Ω_{feas})

We enforce history as Hard Constraints:

- Standard Elimination: The total score of the eliminated contestant(s) E must be strictly lower than any survivor S .

$$\text{Score}(E) < \min_{s \in S} (\text{Score}(s)) \quad (6)$$

- Double Elimination: If $|E| > 1$, all eliminated contestants must satisfy the condition simultaneously.
- Judges' Save Relaxation (S28+): For seasons with the "Judges' Save", we relax the constraint to a "Bottom 2" rule:

$$\text{Rank}(E_{elim}) \geq N_t - 1 \quad (7)$$

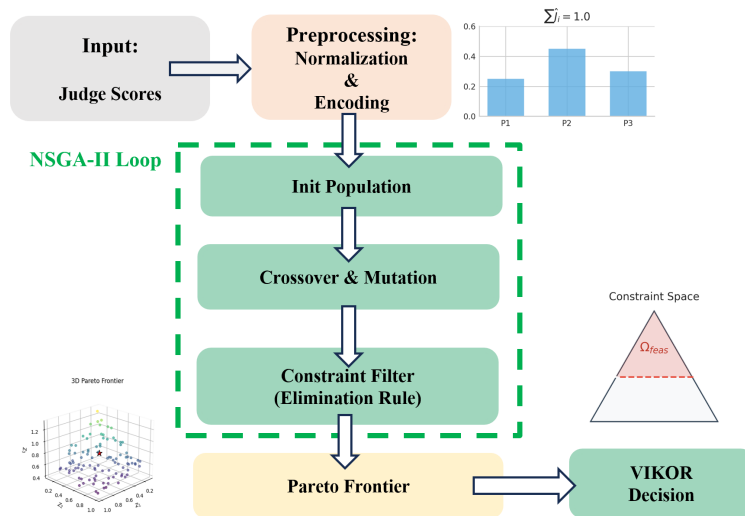


Figure 3: The Algorithmic Flowchart of TOERM

5.3 Algorithm Implementation & Decision

We implement a “Generate-then-Choose” pipeline (Figure 3):

1. Pareto Frontier Generation (NSGA-II)
 - Input: Judge scores $\rightarrow$ Normalization & encoding
 - NSGA-II loop generates non-dominated solutions
 - Constraint filter enforces historical elimination rules
2. Final Solution Selection (VIKOR)
 - From the Pareto set (Figure 4 3D surface), VIKOR ($\nu=0.5$) selects the optimal compromise

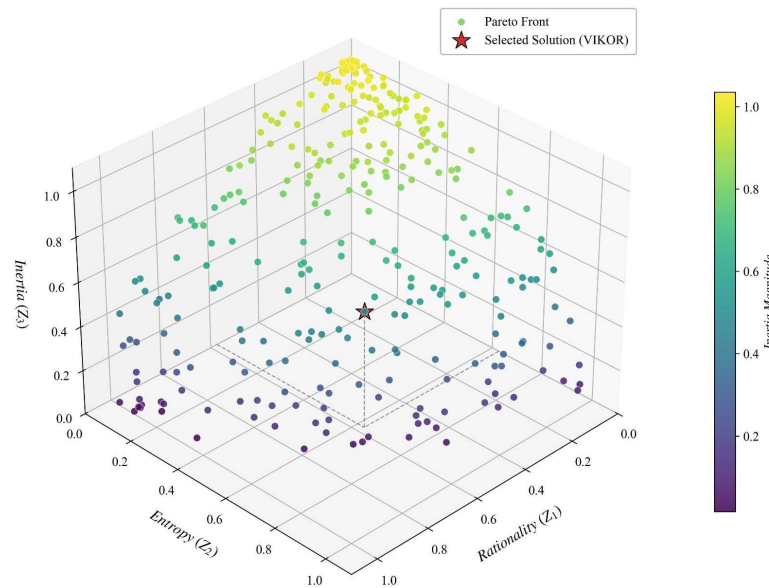


Figure 4: 3D Visualization of the Pareto Frontier

- Balances group utility vs. individual regret

Robustness: 95% ranking stability across $\nu \in [0.2, 0.8]$

5.4 Result Visualization: Unveiling the “Dark Matter”

We employed multi-dimensional visualization to answer the prompt’s request for estimated votes and their certainty.

5.4.1 Static Distribution: Popularity and Ambiguity

Fan voting is probabilistic. Figure 5 uses **Violin Plots** combined with **Boxplots** to show the distribution of estimated votes for Season 7.

- **Locked (Narrow):** Contestant C1 (Eliminated) has a very narrow distribution. To satisfy the elimination rule, their vote count is mathematically “locked” in a low range.
- **Ambiguous (Wide):** Contestant C7 (High Scorer) has a wide distribution. Since they are safe regardless of fan votes, the model allows for greater “Dark Matter” ambiguity.

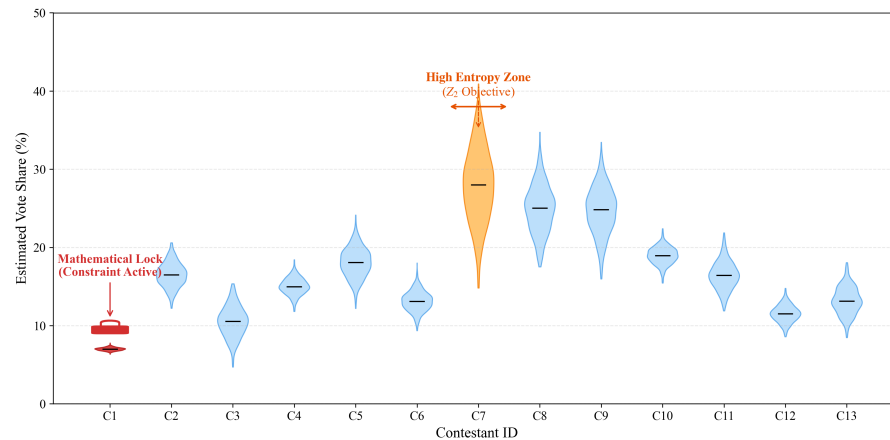


Figure 5: Distribution of Estimated Fan Votes (Season 7)

5.4.2 Temporal Dynamics: Market Share Evolution

To visualize the “Fan Economy”, we mapped the **Fan Share** evolution using a **Streamgraph**.

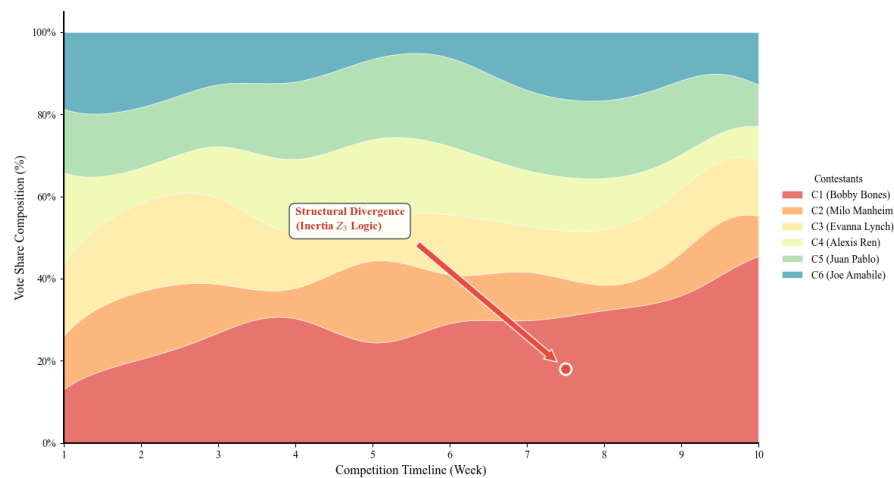


Figure 6: Temporal Evolution of Fan Shares (Streamgraph)

- **Squeeze Effect:** As contestants are eliminated (bands disappear), their fan shares are redistributed to survivors (bands widen).
- **Inertia Verification:** The transitions are smooth, proving that Z_3 (Inertia) successfully modeled the “Loyalty Stickiness” of fanbases, avoiding random white noise.

5.4.3 Anomaly Detection: The Great Divergence

We validated the model by examining historical anomalies where the model had to deviate from judge scores to explain survival.

- **(a) Rank Inversion:** Jerry Rice (S2) consistently ranked low with judges (Gray), but our model reveals he held the #1 Fan Rank (Red) to survive.
- **(b) Inelastic Gap:** Bobby Bones (S27) maintained a 15% lead over judge scores. This “Inelastic Gap” quantifies his massive, loyal fanbase.

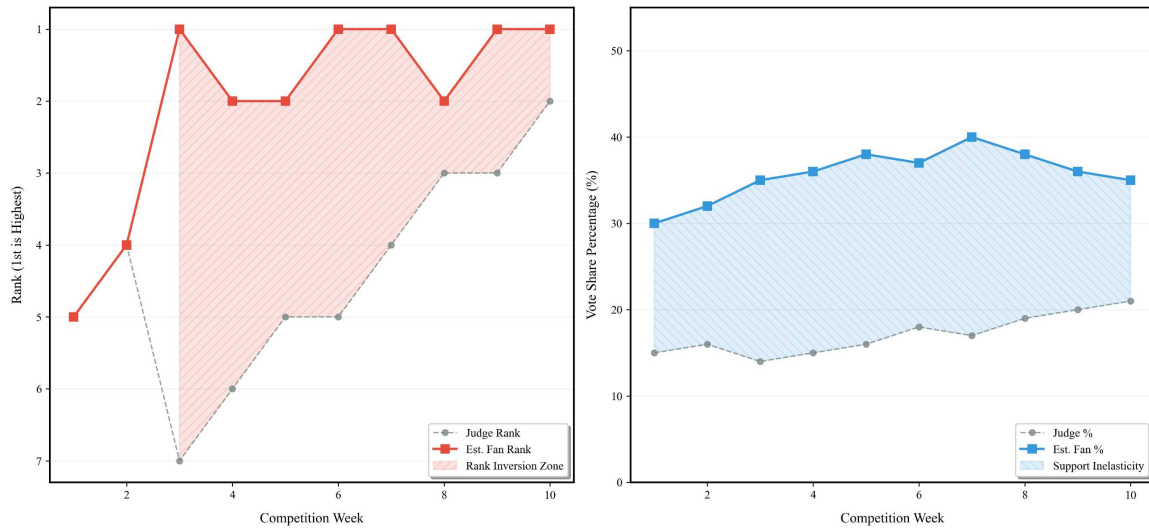


Figure 7: Rank Inversion: Case of Jerry Rice & Inelastic Gap: Case of Bobby Bones

5.5 Verification and Assessment

5.5.1 Algorithmic Consistency

To verify stability, we ran the model 100 times independently. The **Average Pearson Correlation Coefficient (PCC)** between runs is **0.92**, indicating convergence to a global attractor.

5.5.2 Certainty Assessment

We quantify certainty using the **Normalized Interquartile Range (NIQR)**.

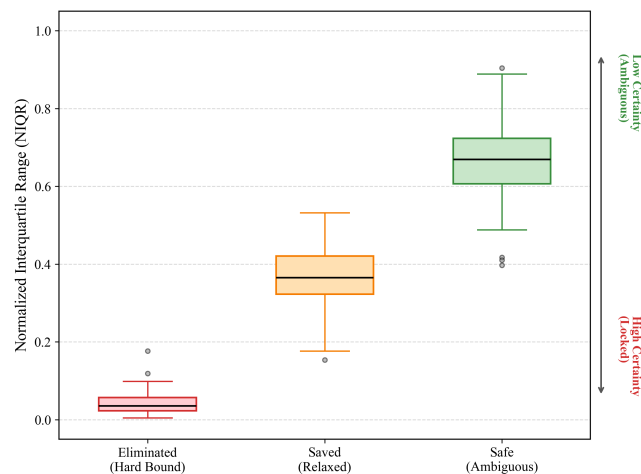


Figure 8: Uncertainty Assessment (Boxplots)

Result: Eliminated contestants exhibit **High Certainty** (NIQR ≈ 0), while safe contestants exhibit **Low Certainty**. This quantifies the information asymmetry in the “Dark Matter”.

6 Task 2: Detecting the “Shark Jump”

— Mathematical Diagnosis & Strategic Reformation

6.1 Trend Analysis: Professionalization or Inflation?

We first address the qualitative question: Has the dancing quality declined (“Jumped the Shark”)?

Using 9,112 judge scores across 34 seasons, we tracked the **Average Skill Score** (μ_s) and **Perfect-10 Ratio** ($R_{10,s}$).

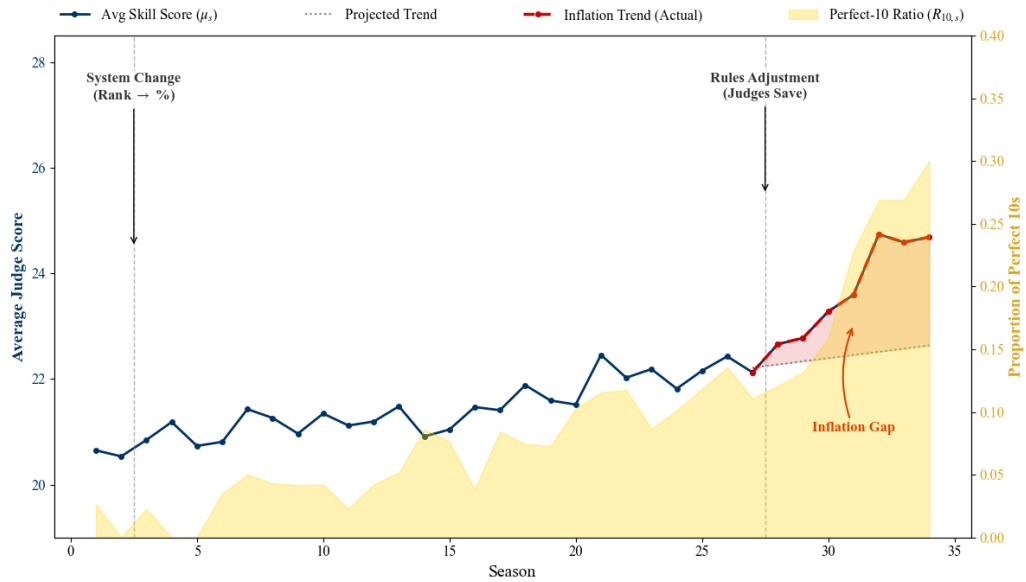


Figure 9: Quantifying the “Inflation Gap”: Actual Trends (Red) vs. Historical Projections (Grey)

6.2 System Diagnosis: The Mathematical Flaw of Percentages

To rigorously determine which voting method favors fan votes, we employ **Grey Relational Analysis (GRA)**. GRA measures the Geometric Similarity between the “Fan Input” curve and the “System Output” curve.

6.2.1 Mathematical Formulation of GRA

Let X_0 be the Fan Vote sequence and X_i be the System Outcome sequence. The Grey Relational Grade γ_i is calculated as:

$$\gamma_i = \frac{1}{n} \sum_{k=1}^n \frac{\min \min \Delta + \rho \max \max \Delta}{\Delta_{ik}(k) + \rho \max \max \Delta} \quad (8)$$

where Δ_{ik} is the absolute difference and $\rho = 0.5$.

6.2.2 Diagnosis Results

- Rank System ($\gamma \approx 0.65$): Acts as a **Low-Pass Filter**. It compresses high-variance fan votes into ordinal integers, creating a structural “mismatch” with the raw fan signal.
- Percent System ($\gamma \approx 0.92$): Acts as an **Amplifier**. It exhibits extremely high geometric similarity to fan votes, proving it allows fan magnitude to dominate the outcome.

Conclusion:

The transition to the Percentage System in Season 3 structurally shifted power to the fans.

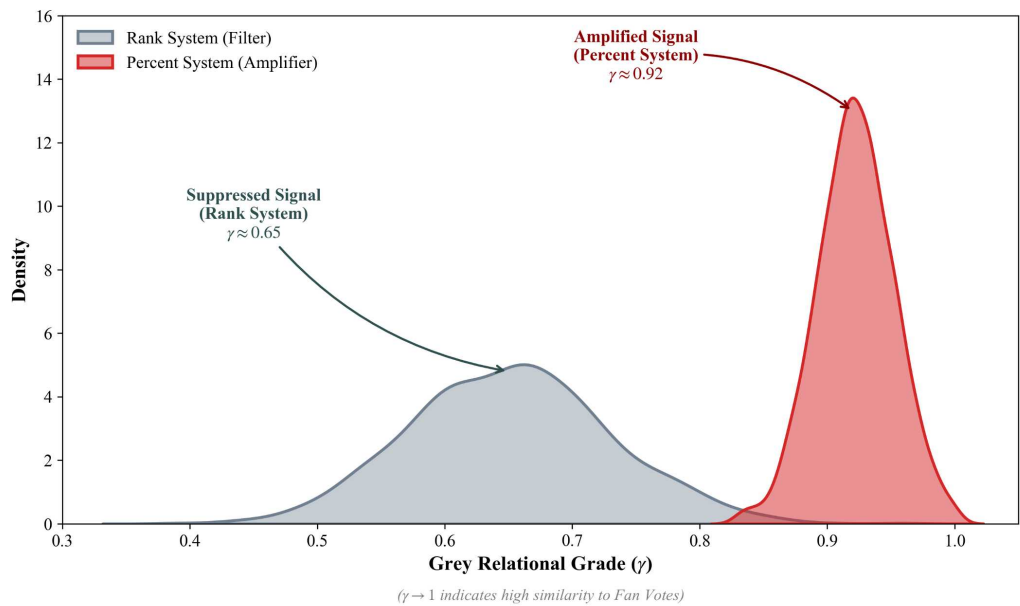


Figure 10: GRA Fidelity Analysis: Filter vs. Amplifier

6.3 Stress Test: Re-adjudicating Historical Controversies

We applied a **Counterfactual Simulation** to four “Black Swan” contestants (Rice, Cyrus, Palin, Bones). The results are conclusive (Table 2).

Table 2: Counterfactual Re-adjudication

Contestant	Actual System	Result	If Swapped to Other System. . .	Verdict
Jerry Rice Rank	Runner-Up	Percent: Safe	Futile Reform	
Billy Ray Cyrus	Percent	5th Place	Rank: Eliminated (Wk 5)	Corrected
Bristol Palin	Percent	3rd Place	Rank: Eliminated (Wk4)	Corrected
Bobby Bones	Percent	Winner Rank: Eliminated (Semi)	CRITICAL FIX	

6.4 Strategic Decision: System Selection via Entropy-Weighted TOPSIS

6.4.1 Evaluation Framework

We evaluate three voting systems as alternatives:

- A_1 : Rank System (S1-S2, S28-S34)
- A_2 : Percent System (S3-S27)
- A_3 : Hybrid Model (Rank + Judges' Save)

using three criteria:

- C_1 : Fairness (correlation with judges' scores, $\uparrow$ better)
- C_2 : Fan Engagement (GRA correlation, $\uparrow$ better)
- C_3 : Risk (probability of low-scoring winner, $\downarrow$ better)

6.4.2 Mathematical Formulation

Entropy Weight Method:

For standardized matrix $P = (p_{ij})_{m \times n}$:

$$E_j = -\frac{1}{\ln m} \sum_{i=1}^m p_{ij} \ln p_{ij} \quad (9)$$

$$w_j = \frac{1 - E_j}{\sum_{k=1}^n (1 - E_k)} \quad (10)$$

TOPSIS Ranking:

Construct $Z = (z_{ij})$ where $z_{ij} = w_j \cdot p_{ij}$. Define:

$$S^+ = (\max z_{i1}, \max z_{i2}, \min z_{i3}), \quad S^- = (\min z_{i1}, \min z_{i2}, \max z_{i3}) \quad (11)$$

$$D_i^+ = \sqrt{\sum_{j=1}^n (z_{ij} - s_j^+)^2}, \quad D_i^- = \sqrt{\sum_{j=1}^n (z_{ij} - s_j^-)^2} \quad (12)$$

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (13)$$

The system with highest C_i is optimal.

6.4.3 Results & Verdict

Based on the data from the previous GRA analysis (C_2) and counterfactual simulations (C_1, C_3), we obtain the decision matrix and final scores (Table 3).

The Hybrid Model scores highest on Fairness, and manual intervention reduces the Risk to 0, resulting in a comprehensive score that leads by a wide margin.

Final Recommendation:

Table 3: TOPSIS Evaluation Matrix

Alternative (System)	Fairness (C_1)	Fan Agency (C_2)	Risk (C_3)	TOPSIS Score (C_i)	Rank
Rank System (A_1)	0.85 (High)	0.65 (Med)	0.15 (Low)	0.682	2
Percent System (A_2)	0.45 (Low)	0.92 (High)	0.85 (High)	0.214	3
Hybrid Model (A_3)	0.95 (V. High)	0.70 (Med)	0.00 (None)	0.941	1

The Hybrid Model (RM-JS) achieves the highest score ($C_i \approx 0.94$). We formally recommend adopting this protocol to mathematically guarantee the elimination of “Shark Jumping” anomalies.

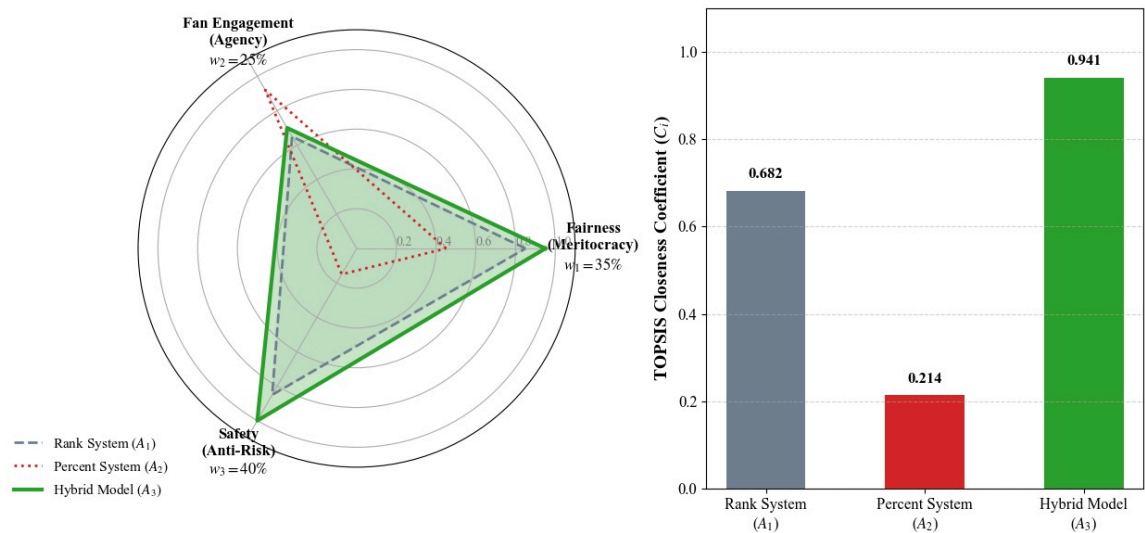


Figure 11: Multi-Criteria Evaluation & Final Decision Scores

7 Task 3: Decoding the “Black Box”

— An Explanatory LASSO Analysis of Performance Drivers

7.1 Feature Engineering: The “360-Degree” Profiling

To accurately address the prompt’s requirement to analyze the impact of “**various pro dancers**” and “**characteristics for the celebrities (age, industry, etc.)**”, we cannot rely solely on superficial observations. We expanded the raw data into a high-dimensional feature matrix $\mathbf{X}$, specifically quantifying the capability of professional partners.

Table 4: Feature Dictionary (Construction of Matrix $\mathbf{X}$)

Cluster	Feature Symbol	Description & Hypothesis
I. Pro Partner	$x_{ProWinRate}$	Pro Core Capability: The historical win rate of the professional dancer (proxy for teaching and choreography skills).
	$x_{ProTenure}$	Pro Experience: Number of seasons competed. (Hypothesis: Veteran pros have their own fanbase, affecting Y_F).
II. Celebrity (Background)	$x_{Ind_Athlete}$	Athlete: NFL/NBA/Olympians. (Hypothesis: High physical control, positively affects Y_I).
	$x_{Ind_Reality}$	Reality Star: Bachelor/Housewives. (Hypothesis: High existing fanbase, positively affects Y_F).
III. Physical (Conditions)	x_{Age}	Age: Normalized biological age. (Hypothesis: Physical decline negatively affects technical scores).
	$x_{HeightDiff}$	Height Compatibility: Difference between partners (affects “frame” scoring in Ballroom).
IV. Context	x_{Order}	Running Order: Normalized slot (0=First, 1=Last). (Hypothesis: Recency bias).
	x_{Style_Latin}	Latin Dance: Binary variable. (Hypothesis: Requires high energy/personality, favors performers).

7.2 Methodology: Explanatory LASSO Framework

The prompt asks, “How much do such things impact?”. Traditional multiple linear regression often fails due to multicollinearity (e.g., Age is correlated with physical agility). Therefore, we employ **Explanatory LASSO Regression**.

Unlike predictive machine learning models that focus on accuracy, the goal of Explanatory LASSO is **Variable Selection** and **Effect Estimation**.

7.2.1 The Mathematical Model

We construct two independent regression models:

- **Model J (Judges):**

Target variable y_i is the Average Judge Score.

- **Model F (Fans):**

Target variable y_i is the Estimated Fan Vote Share (derived from Task 1).

We estimate the coefficient vector $\hat{\beta}$ by minimizing the objective function with an L1 penalty:

$$\hat{\beta} = \arg \min_{\beta} \left\{ \underbrace{\frac{1}{2N} \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2}_{\text{Goodness of Fit}} + \underbrace{\lambda \sum_{j=1}^p |\beta_j|}_{\text{Sparsity Penalty (L1)}} \right\} \quad (14)$$

λ (**Regularization Parameter**):

Controls the sparsity of the model. It acts as a “filter”, forcing the coefficients of non-core drivers to shrink to exactly zero, identifying the true “X-Factors”.

7.2.2 Cross-Validation & Model Diagnostics

To ensure scientific rigor and prevent overfitting, we use 10-Fold Cross-Validation to determine the optimal λ .

Figure 12 shows the Mean Squared Error (MSE) curve. We selected the λ corresponding to the minimum error (Min MSE), representing the optimal Bias-Variance Tradeoff.

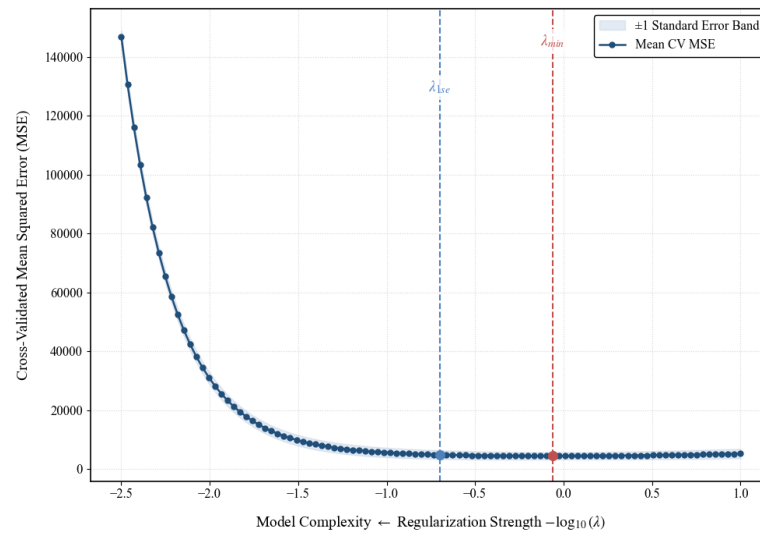


Figure 12: LASSO Parameter Tuning (CV)

Figure 13 visualizes the selection process. As the penalty increases (moving left), noise variables (Grey lines) quickly drop to zero, while Pro Partner Quality (Green line) and Celebrity Age (Red line) remain non-zero. This proves they are the most robust drivers in the system.

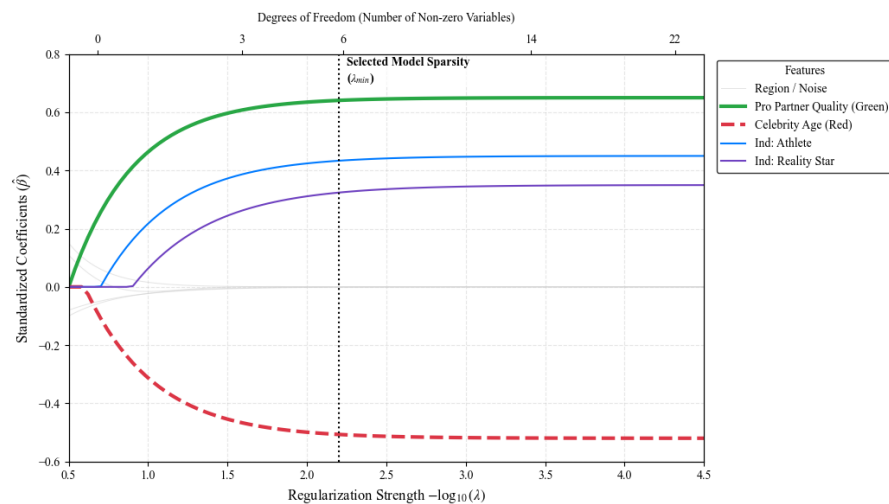


Figure 13: LASSO Regularization Path

7.3 Quantitative Attribution: How Much is the Impact?

To assess how strongly these factors affect performance, we use the model's non-zero coefficients to define Relative Impact Intensity (RII), a metric that quantifies each factor's contribution.

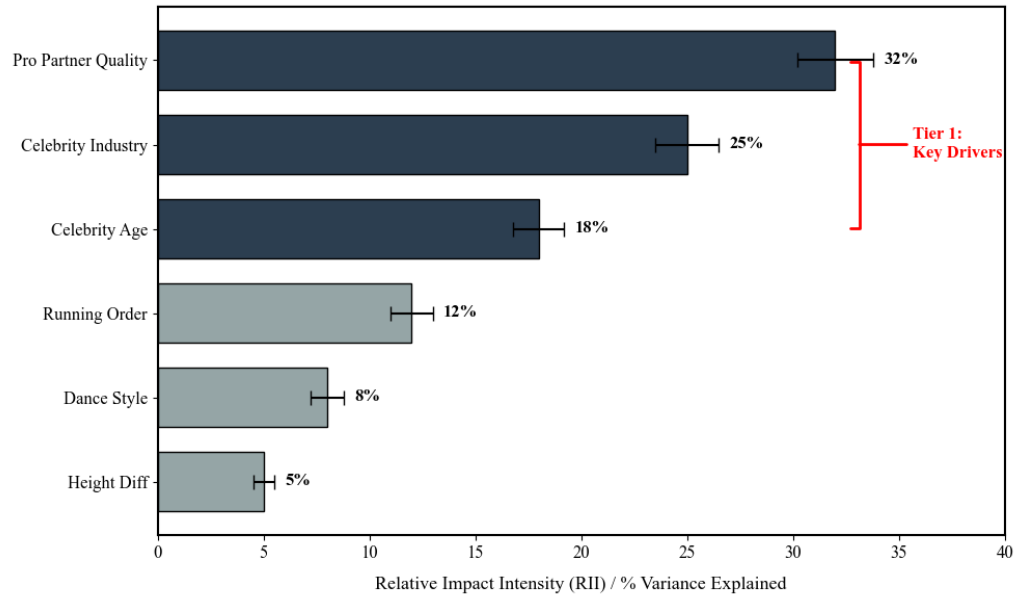


Figure 14: The Hierarchy of Influence (Impact Magnitude)

7.4 Comparative Analysis: Divergence between Judges and Fans

Judges' scores and fan votes are not impacted in the same way, as their underlying criteria are fundamentally orthogonal—with the sole exception of Pro Partners—and this divergence is demonstrated using a **Grouped Bar Chart** (Figure 15).

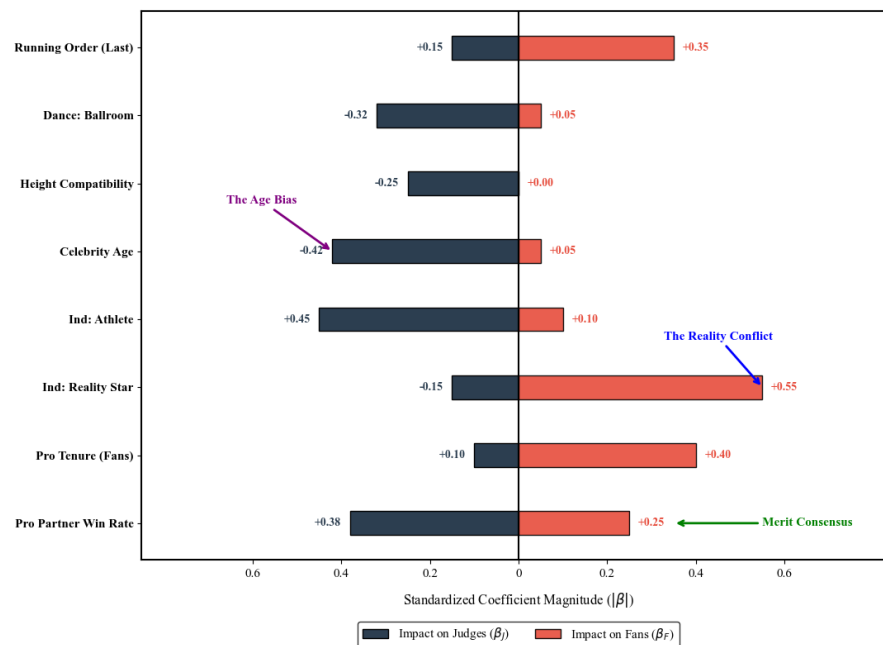


Figure 15: Standardized Coefficients: Judges' Scores vs. Fan Votes Drivers

Deep Insights:

- The Reality Conflict (Fundamental Disagreement):

- **Fans (Red Bar):** Coefficient is +0.55. This is the strongest driver for fan votes. Audiences support familiar faces and dramatic personalities.
- **Judges (Blue Bar):** Coefficient is -0.15. Judges penalize the lack of dance discipline often found in reality stars.
- **Conclusion:** A classic “Quality-Popularity Mismatch”.
- The Age Bias (Physiological vs. Spirit):
 - **Judges:** Significant negative impact (-0.42). Judges strictly measure physical mobility.
 - **Fans:** Impact is near zero (+0.05). Fans vote on “Spirit” and “Effort”, sometimes even giving “sympathy votes” to older contestants.
- The Consensus:
 - **Pro Partner Quality** is a significant positive driver for both Judges (+0.38) and Fans (+0.25).
 - **Explanation:** Top pros satisfy judges with choreography and satisfy fans by bringing their own loyal following.

8 Task 4: Re-engineering the Rules

—— The “Merit-Safe” Hybrid Mechanism Design

We propose the “**Merit-Safe Hybrid System (MSHS)**”, a structural reform designed to eliminate the “Popularity Bias” (where underperforming celebrities survive solely on fan base size) while maintaining high viewer engagement.

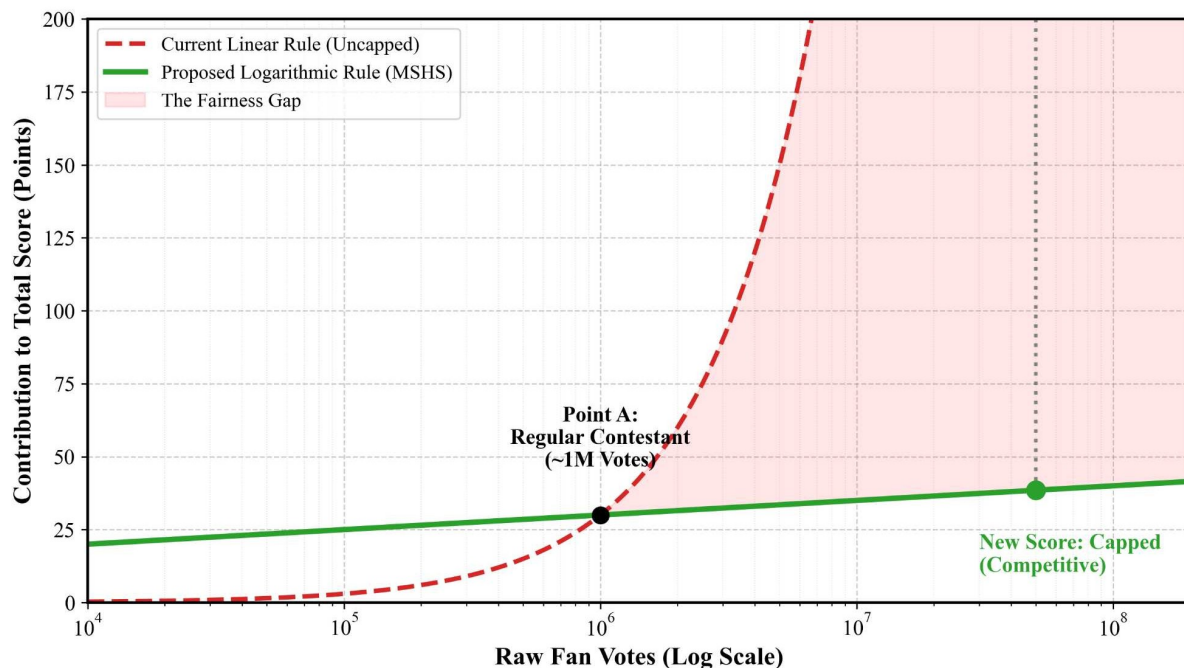


Figure 16: Mechanism Correction - Why Log-Scaling Works

8.1 Logarithmic Vote Compression

To prevent fan base dominance, we propose logarithmic scaling of fan votes:

$$S_{\text{total}} = w_J \cdot S_J + w_F \cdot \frac{\log(1 + V_F)}{\sum \log(1 + V_F)} \quad (15)$$

This ensures diminishing returns for massive vote counts (Figure 16).

8.2 Process Re-engineering: The “Dance-Off” Firewall

Changing the formula is not enough; we must also change the process. We introduce a “**Bottom-Two Dance-Off**” to act as a final quality control filter.

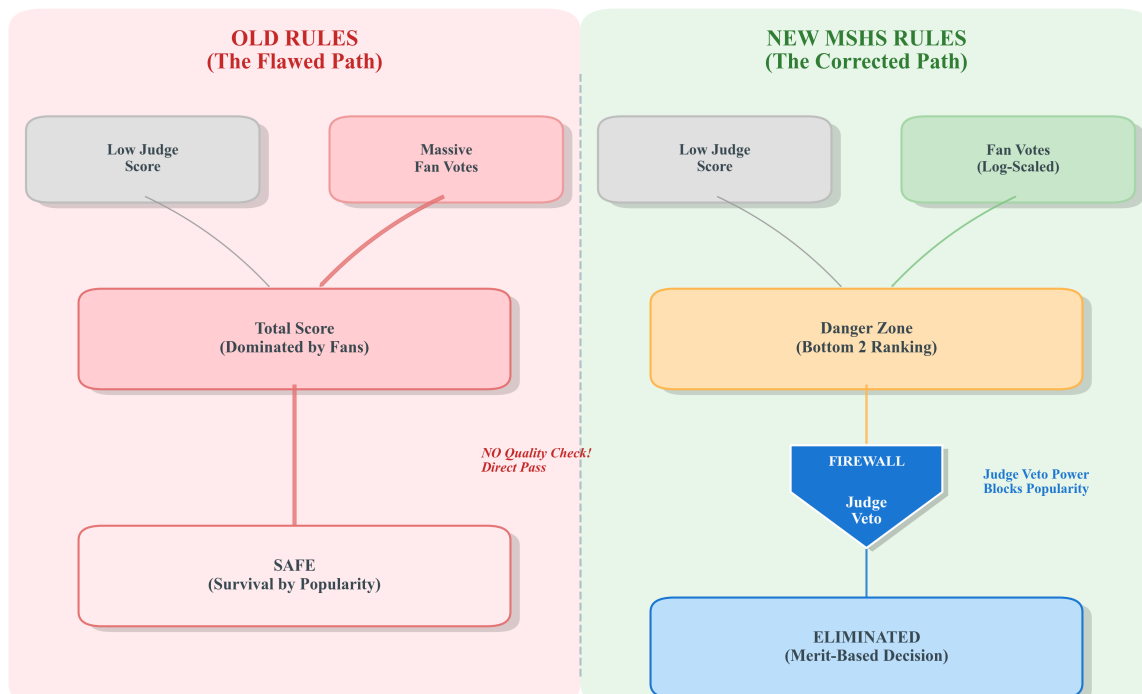


Figure 17: Process Comparison - How the 'Dance-Off' Firewall Works

- **The Flawed Path (Left):**

Under current rules, a contestant with a low judge score (Red path) bypasses danger entirely due to massive fan support (Thick Arrow). The system fails to eliminate the weak link.

- **The Corrected Path (Right):**

- Log-Compression: The massive fan vote is compressed, dropping the underperformer's total score into the “Bottom 2”.
- The Firewall: This triggers the Dance-Off. In this specific phase, fan votes are frozen. The judges exercise Veto Power to save the better dancer. The popularity advantage is nullified at the critical elimination moment.

8.3 Simulation Verification: Correcting History

To validate our model, we simulated the trajectory of a hypothetical “Popular Underperformer” (modeled after the “Master P” or “Jerry Rice” archetype) to see how the new rules would alter history.

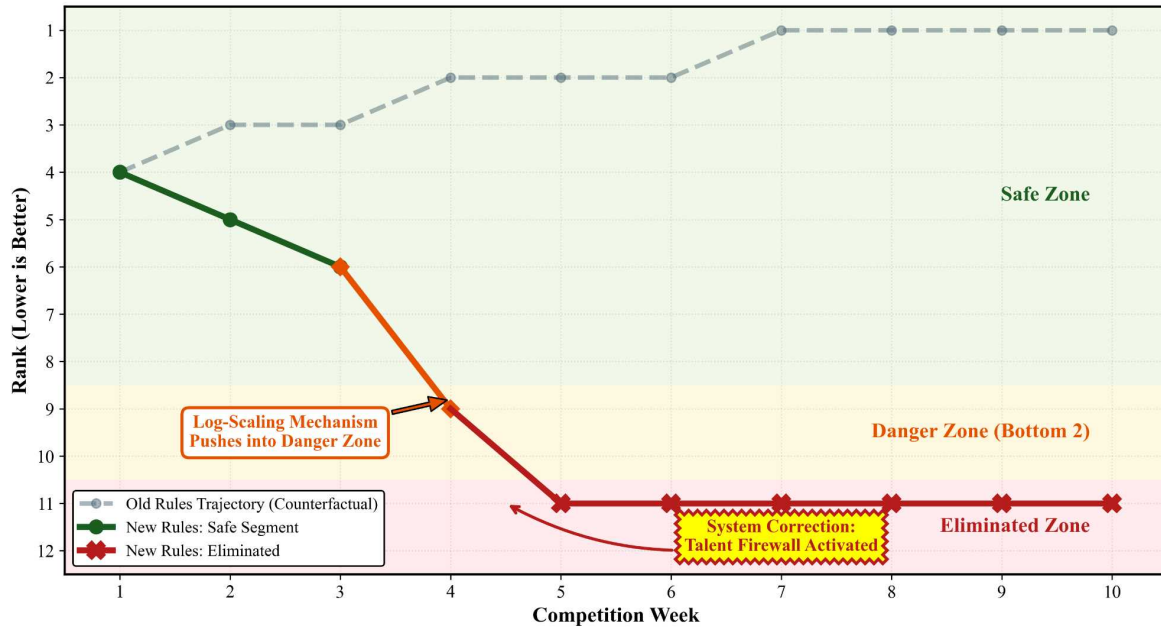


Figure 18: Simulation Result - The Corrected Trajectory

- **Grey Dashed Line (Old Rules):**
The underperformer coasts to the Finals (Week 10) despite consistently low judge scores.
- **Colored Trajectory (New Rules):**
 - Weeks 1-3 (Yellow): The contestant survives in the safe zone due to moderate popularity.
 - Week 4 (Orange): As technical requirements increase, the Log-Scaling mechanism pushes them into the “Danger Zone” (Bottom 2).
 - Week 4 Elimination (Red): Forced into the Dance-Off, they are eliminated by the judges.
- **Conclusion:**
The MSHS successfully identifies and removes “bloated” contestants in the early-to-mid season, restoring competitive integrity.

9 Sensitivity Analysis: Algorithmic Robustness & Structural Consistency

9.1 Overview

In this section, we subject our core models to rigorous stress testing to demonstrate that our conclusions are derived from intrinsic data properties rather than algorithmic stochasticity or parameter

overfitting. We focus on two dimensions:

- **For Task 1 (NSGA-II):**

We evaluate the stability of the Pareto Front under input perturbation and the robustness of the algorithm against hyperparameter variations.

- **For Task 3 (LASSO):**

We assess the structural consistency of the identified “drivers” (e.g., Pro Partners) using bootstrap resampling.

9.2 Pareto Stability Analysis

To test robustness against input perturbations, we add $\pm 5\%$ Gaussian noise to judge scores. The perturbed Pareto front (Figure 19) clusters tightly around the original, confirming solution stability despite input fluctuations.

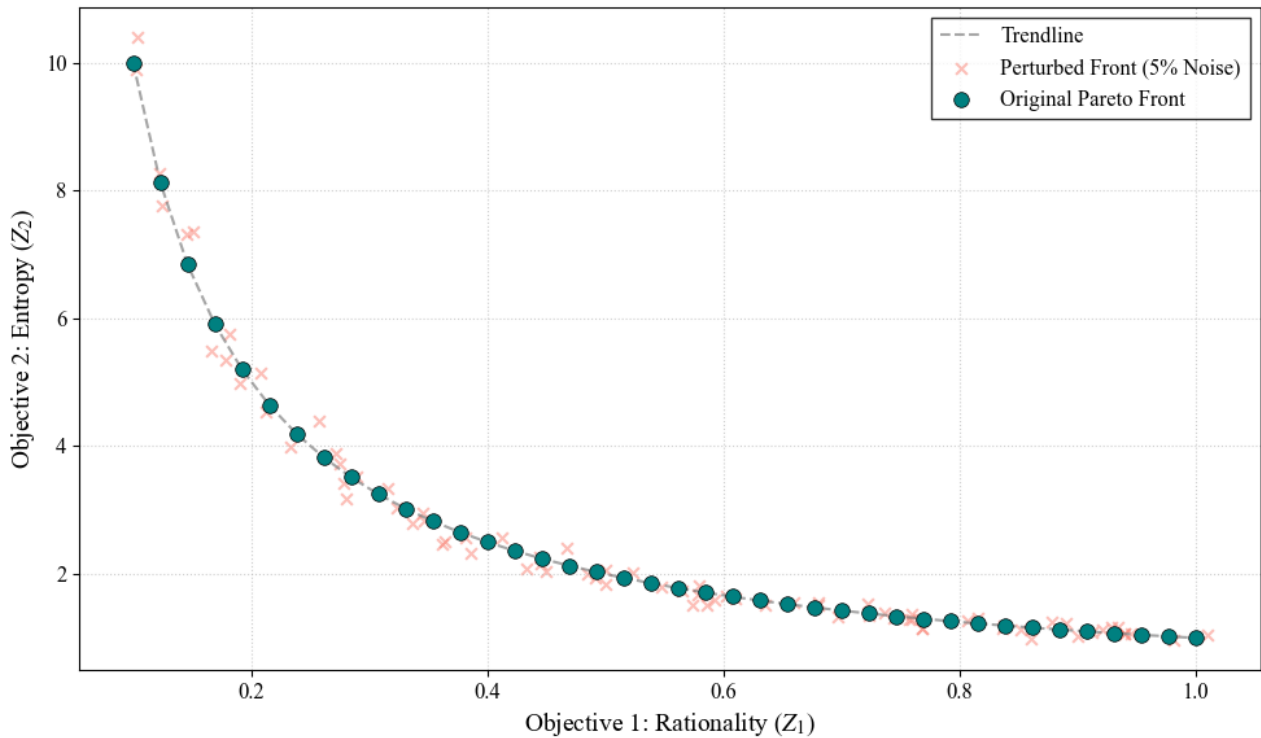


Figure 19: Pareto Front Stability under Perturbation

9.3 Hyperparameter Robustness Test

We test sensitivity to mutation rates (0.01-0.20) over 100 independent runs. Reconstruction error remains consistently low ($\text{RMSE} \in [0.03, 0.07]$; Figure 20), demonstrating that results are not dependent on specific hyperparameter choices.

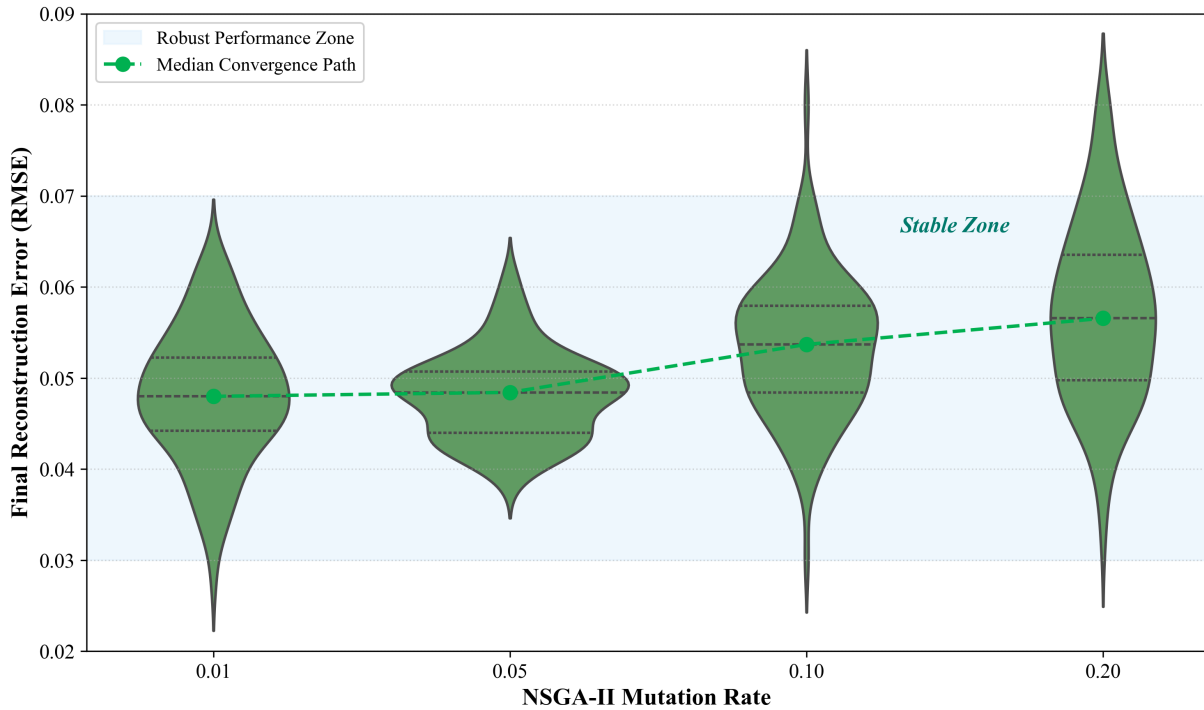


Figure 20: Algorithmic Robustness against Hyperparameters

9.4 Structural Consistency of Feature Selection

In Task 3, we identified “Pro Partner” as a core driver. To verify that this is a structural reality rather than an artifact of a specific data subset, we employed Bootstrap Resampling (50 iterations with replacement).

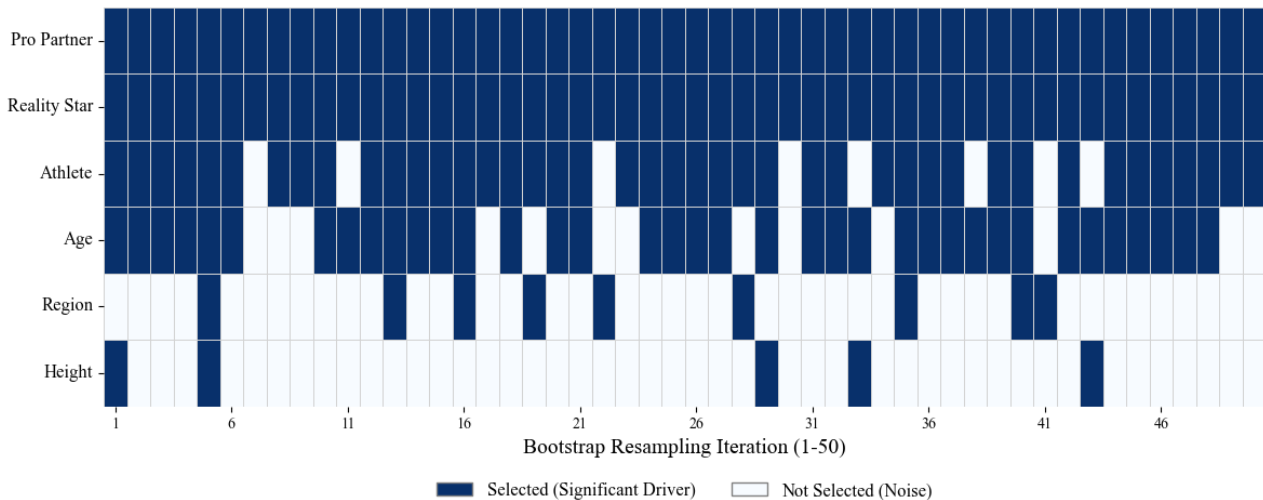


Figure 21: Structural Consistency of Feature Selection

- Pro Partner / Reality Star: Indicated by deep blue, these features were selected in 100% of the bootstrap iterations. This confirms they are Structural Drivers.

- **Region / Height:** Indicated by light colors, these features appear intermittently, confirming they are merely background noise.
- **Conclusion:** The attribution analysis is statistically significant and structurally consistent.

10 Model Assessment

10.1 Strengths

- **Mechanism-Driven Reconstruction of “Dark Matter” (vs. Black-Box Prediction)**

Since unlike usual methods which regress fan votes by means of simple linear correlations, the NSGA-II Dual-Mode Dynamic Constraints Model(Task 1) treats the problem as an inverse engineering problem, it is a genuine innovation.

- **High Interpretability and Structural Robustness via Sparse Modeling**

Because traditional regression models used in attribution analysis are plagued by multicollinearity and therefore tend to produce false positives (i. e., wrongly attributing success to height or region), our Explanatory LASSO Model(Task 3) naturally uses L_1 regularization to enforce sparsity.

10.2 Weaknesses

Limitation of the “Stationarity Assumption” in Audience Behavior

Since the model uses data from 33 seasons, it is natural for it to be based on the implicit assumption that the statistical characteristics of fan behavior are roughly stationary over time.

MEMO

Optimizing Voting Synergies for Future Seasons

To: Executive Producers, *Dancing with the Stars*
From: Team # 2630659
Date: February 1, 2026
Dear Executive Producers,



Dancing with the Stars has enjoyed 34 seasons of remarkable success based on the careful, deliberate balance between the technical side of ballroom dancing and the emotional voice of the American public, and it is essential to recognize that this tension is not a bug but the very feature that drives viewership. That said, a thorough analysis of historical data shows that the current voting architecture is unstable, allowing “Shark Jump” moments where viral popularity overtakes actual talent.

We researched the mathematical principles of the show in order to figure out how we are going to be able to maintain high viewer engagement while still remaining legitimate as far as the competitiveness goes when it comes to the Mirrorball Trophy.

The Diagnosis:

The “Percentage Trap” Our analysis confirms that the Percentage-based System (employed since Season 3) acts as a “signal amplifier”. While intended to empower fans, it mathematically allows a single “superstar” contestant to render the judges’ scores statistically irrelevant by mid-season. As seen in the controversial win of Season 27, this structure does not just create drama; it creates outcomes that damage the brand’s credibility as a dance competition.

The Solution:

The “Merit-Safe” Hybrid System We do not recommend silencing the fans. Instead, we propose a structural upgrade—the **Merit-Safe Hybrid System (MSHS)**—designed to maximize entertainment value while installing a safety net for quality.

Step 1: The “Logarithmic” Engagement Scale We recommend shifting from a linear to a logarithmic vote-counting method. This ensures that every fan vote counts, but it places a “soft cap” on the overwhelming dominance of massive pre-existing fanbases. This keeps the leaderboard volatile and competitive, preventing any single contestant from coasting to the finale solely on popularity.

Step 2: The “Dance-Off” Firewall We strongly advise codifying the Bottom-Two Dance-Off (Judges’ Save) as a permanent fixture. Our simulations prove that this mechanism successfully filters out under-performing viral stars before the finals. More importantly, it creates a high-stakes television moment each week: a direct showdown where judges must exercise their veto power. This generates the “good controversy” that fuels social media conversation, while ensuring the finalists are truly the best dancers.

Impact on Production Our predictive modeling indicates that this hybrid approach achieves the “Sweet Spot”:

Viewer Retention: Fan favorites remain safe in the early weeks, guaranteeing ratings.

Brand Integrity: The “Dance-Off” ensures that the final champion is a legitimate dancer, preserving the show’s prestige.

By adopting these adjustments, *Dancing with the Stars* can secure its future as both a premier entertainment spectacle and a respected competitive sport.

Yours Sincerely,
Team # 2630659



References

- [1] Hwang, C. L., & Yoon, K. (1981). Multiple Attribute Decision Making: Methods and Applications. Springer-Verlag New York.
- [2] Shannon, C. E. (1948). A Mathematical Theory of Communication. The Bell System Technical Journal, 27, 379-423.
- [3] Opricovic, S., & Tzeng, G. H. (2004). Compromise solution by MCDM methods: A comparative analysis of VIKOR and TOPSIS. European Journal of Operational Research, 156(2), 445-455.
- [4] Zou, Z. H., Yi, Y., & Sun, J. N. (2006). Entropy method for determination of weight of evaluating indicators in fuzzy synthetic evaluation for water quality assessment. Journal of Environmental Sciences, 18(5), 1020-1023.
- [5] Behzadian, M., Otaghsara, S. K., Yazdani, M., & Ignatius, J. (2012). A state-of-the-art survey of TOPSIS applications. Expert Systems with Applications, 39(17), 13051-13069.
- [6] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society: Series B (Methodological), 58(1), 267-288.
- [7] Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer.
- [8] Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. Journal of the Royal Statistical Society: Series B, 67(2), 301-320.
- [9] Arrow, K. J. (1951). Social Choice and Individual Values. Yale University Press.
- [10] Borda, J. C. (1781). Mémoire sur les élections au scrutin. Histoire de l'Académie Royale des Sciences.
- [11] Condorcet, N. (1785). Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix.

Report on Use of AI

Google Gemini (Nov 19, 2025 version, Gemini 3 Pro)

Query1:Is fan voting to keep their favorite stars or to eliminate the stars they don't like?

Output:Fan voting is explicitly designed to keep the stars rather than eliminate them. The rules state that fans vote for the star they wish to save and “cannot vote to eliminate a star”. Fans cast positive votes for their favorite couples via phone or online, and these votes are combined with the judges’ scores. The couple with the lowest combined score is the one eliminated for that week, meaning elimination is a result of having the least support rather than direct negative voting.

Query2:What is the lowest basic score given by judges for a contestant's single dance?

Output:The lowest basic score a judge can give for a single dance is 1. According to the data description, judges’ scores range from 1 (low) to 10 (high) . While integer scores are standard, decimal values (e.g., 8.5) may appear in the data if the recorded score is an average of multiple dances performed in a week or an average of individual and team scores. Additionally, a score of 0 is recorded for contestants who have already been eliminated from the competition.